

120 Days of Quantitative Finance

Week 15: Advanced Time Series Modelling

Day 103: GARCH Models and Volatility Clustering

1 Introduction

Financial time series often exhibit **volatility clustering**, where periods of high volatility are followed by high volatility, and low volatility follows low volatility.

GARCH models are designed to capture this behavior.

2 ARCH Model

The ARCH(q) model defines conditional variance as:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2$$

3 GARCH Model

The GARCH(p, q) model extends ARCH:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

4 GARCH(1,1)

The most commonly used model:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

5 Interpretation

- α_1 captures impact of recent shocks
- β_1 captures persistence of volatility
- High $\alpha_1 + \beta_1$ implies strong clustering

6 Volatility Clustering

- large shocks tend to be followed by large shocks
- small shocks tend to be followed by small shocks

7 Financial Relevance

GARCH models are used for:

- volatility forecasting
- risk management (VaR)
- option pricing

8 Conclusion

GARCH models provide a framework to model time-varying volatility, which is a key feature of financial markets.